

Scenario 2: Critical Healthcare Dashboard

Context:

You are tasked with building a **Critical Care Dashboard** for a hospital network. The system integrates with **IoT devices** (e.g., heart rate monitors, ventilators) and provides real-time health metrics to doctors and nurses. The dashboard must alert staff about critical issues like low oxygen levels or heart irregularities, enabling immediate intervention.

Key stakeholders have conflicting requirements:

- **Doctors:** Need instant alerts and smooth access to the dashboard, especially during emergencies. They often move between patients and demand a single login session across devices.
- **Compliance Team:** Requires adherence to stringent data privacy laws (e.g., HIPAA) and demands that every user session be secure, with detailed audit trails for any data access.
- **IT Department:** Warns that implementing strong security features (e.g., multi-factor authentication, session timeouts) could slow down system interactions. They also point out that building a flexible system to accommodate different departments will increase development time.
- **Hospital Administrators:** Insist that the system must be delivered within **six months** to replace the aging legacy system.

Additional constraints:

1. The system must run on limited on-premise infrastructure with fallback options in case of power or network failures.
2. The budget is tight, and the hospital cannot afford to hire additional developers or invest in expensive cloud services.

Challenge for Architects:

1. What tradeoffs need to be considered?
 2. How would you balance the need for robust security with ease of use for doctors in emergencies?
 3. What would your architecture look like (e.g., session management, alerting mechanisms, data access)? Justify your design decisions.
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